

Axcel®

AZITHROMYCIN-250mg TABLET

COMPOSITION:

Each tablet contains
Azithromycin Dihydrate
equivalent to Azithromycin

PRESENTATION:

White to off white colour, biconvex, oval shape tablet, embossed Kotra logo on one side and plain on the other side.

INDICATIONS:

Azithromycin is indicated for infections caused by susceptible organisms; in lower respiratory tract infections including bronchitis and pneumonia, in skin and soft tissue infections, in acute otitis media and in upper respiratory tract infections including sinusitis and pharyngitis / tonsillitis. (Penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including the prophylaxis of rheumatic fever. Azithromycin is generally effective in the eradication of streptococci from the oropharynx; however, data establishing the efficacy of azithromycin and the subsequent prevention of rheumatic fever are not available at present). In sexually transmitted diseases in men and women, azithromycin is indicated for the treatment of uncomplicated genital infections due to *Chlamydia trachomatis*. It is also indicated for the treatment of chancroid due to *Haemophilus ducreyi* and uncomplicated genital infections due to non-multiresistant *Neisseria gonorrhoeae*; concurrent infection with *Treponema pallidum* should be excluded.

PHARMACOLOGY:

Mechanism of Action:

Pharmacotherapeutic group: antibacterials for systemic use; macrolides; azithromycin, ATC code J01FA.

Mode of action:

Azithromycin is an azalide, a sub-class of the macrolid antibiotics. By binding to the 50S-ribosomal sub-unit, azithromycin avoids the translocation of peptide chains from one side of the ribosome to the other. As a consequence of this, RNA-dependent protein synthesis in sensitive organisms is prevented.

PK / PD relationship:

For azithromycin the AUC / MIC is the major PK/PD parameter correlating best with the efficacy of azithromycin. Following the assessment of studies conducted in children, the use of azithromycin is not recommended for the treatment of malaria, neither as monotherapy nor combined with chloroquine or artemisinin based drugs, as noninferiority to anti-malarial drugs recommended in the treatment of uncomplicated malaria was not established.

Mechanism of resistance:

Resistance to azithromycin may be inherent or acquired. There are three main mechanisms of resistance in bacteria: target site alteration, alteration in antibiotic transport and modification of the antibiotic. Complete cross resistance exists among *Streptococcus pneumoniae*, beta-haemolytic streptococcus of group A, *Enterococcus faecalis* and *Staphylococcus aureus*, including methicillin resistant *S. aureus* (MRSA) to erythromycin, other macrolides and lincosamides.

Breakpoints:

EUCAST (European Committee on Antimicrobial Susceptibility Testing)

	Susceptible (mg/l)	Resistant (mg/l)
<i>Staphylococcus spp.</i>	≤ 1	> 2
<i>Streptococcus spp.</i>	≤ 0.25	> 0.5
<i>Streptococcus pneumoniae</i>	≤ 0.25	> 0.5
<i>Haemophilus influenzae</i>	≤ 0.12	> 4
<i>Moraxella catarrhalis</i>	≤ 0.5	> 0.5
<i>Neisseria gonorrhoeae</i>	≤ 0.25	> 0.5

Susceptibility:

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable. Pathogens for which resistance may be a problem : prevalence of resistance is equal to or greater than 10% in at least one country in the European Union.

Commonly susceptible species:

Aerobic Gram-negative microorganisms : *Haemophilus influenzae**, *Moraxella catarrhalis**
Other microorganisms : *Chlamydia pneumonia*, *Chlamydia trachomatis*, *Legionella pneumophila*, *Mycobacterium avium*, *Mycoplasma pneumonia**

Species for which acquired resistance may be a problem:

Aerobic Gram-positive microorganisms : *Staphylococcus aureus**, *Streptococcus agalactiae*, *Streptococcus pneumoniae**, *Streptococcus pyogenes**
Other microorganisms : *Ureaplasma urealyticum*

Inherently resistant organisms:

Aerobic Gram-positive microorganisms :
Staphylococcus aureus – methicillin resistant and erythromycin resistant strains
Streptococcus pneumoniae – penicillin resistant strains
Aerobic Gram-negative microorganisms : *Escherichia coli*, *Pseudomonas aeruginosa*, *Klebsiella spp.*
Anaerobic Gram-negative microorganisms : *Bacteroides fragilis* group
*Clinical effectiveness is demonstrated by sensitive isolated organisms for approved clinical indications

Pharmacokinetics:

Absorption:

After oral administration the bioavailability of azithromycin is approximately 37%. Peak plasma levels are reached after 2-3 hours (C_{max} after a single dose of 500mg orally was approximately 0.4mg/l).

Distribution:

Kinetic studies have shown markedly higher azithromycin levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the active substance is heavily tissue bound (steady state distribution volume of approximately 311kg). Concentrations in target tissues such as lung, tonsil, and prostate exceed the MIC₅₀ for likely pathogens after a single dose of 500mg. In experimental *in vitro* and *in vivo* studies azithromycin accumulates in the phagocytes, freeing is stimulated by active phagocytosis. In animal studies this process appeared to contribute to the accumulation of azithromycin in the tissue. In serum the protein binding of azithromycin is variable and depending on the serum concentration varies from 50% in 0.05mg/l to 12% in 0.5mg/l.

Excretion:

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. About 12% of an intravenously administered dose is excreted in the urine unchanged over a period of 3 days; the majority in the first 24 hours. Biliary excretion of azithromycin, predominantly in unchanged form, is a major route of elimination. The identified metabolites (formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by the splitting of the dadione conjugate) are microbiologically inactive. After a 5 day treatment slightly higher (29%) AUC values were seen in the elderly volunteers (> 65 years of age) compared to the younger volunteers (< 45 years of age). However, these differences are not regarded as clinically relevant; therefore a dose adjustment is not recommended.

Pharmacokinetics in special populations:

Elderly:

The pharmacokinetics of azithromycin in elderly men was similar to that of young adults; however, in elderly women, although higher peak concentrations (increased by 30-50%) were observed, no significant accumulation occurred.

Renal insufficiency:

Following a single oral dose of azithromycin 1g, mean C_{max} and AUC₀₋₁₂₀ increased by 5.1% and 4.2% respectively, in subjects with mild to moderate renal impairment (glomerular filtration rate of 10-80ml/min)

compared with normal renal function (GFR > 80ml/min). In subjects with severe renal impairment, the mean C_{max} and AUC₀₋₁₂₀ increased 61% and 33% respectively compared to normal.

Hepatic insufficiency:

In patients with mild to moderate hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to normal hepatic function. In these patients, urinary recovery of azithromycin appears to increase perhaps to compensate for reduced hepatic clearance.

Infants, toddlers, children and adolescents:

Pharmacokinetics have been studied in children aged 4 months – 15 years taking capsules, granules or suspension. At 10mg/kg on day 1 followed by 5mg/kg on days 2-5, the C_{max} achieved is slightly lower than adults with 224µg/l in children aged 0.6-5 years and after 3 days dosing and 383µg/l in those aged 6-15 years. The t_{1/2} of 36h in the older children was within the expected range for adults.

DOSAGE AND ADMINISTRATION:

For oral administration only: Axel Azithromycin-250mg Tablet should be administered as a single daily dose. The period of dosing with regard to infection is given below. It can be taken with or without food.

Adults:

For the treatment of sexually transmitted diseases caused by *Chlamydia trachomatis* and *Haemophilus ducreyi*, the dose is 1000mg as a single oral dose. For susceptible *Neisseria gonorrhoeae*, the dose is 1000mg or 2000mg of azithromycin in combination with 250mg or 500mg ceftriaxone according to local clinical treatment guidelines. For patients who are allergic to penicillin and / or cephalosporins, prescribers should consult local treatment guidelines. For all other indications in which the oral formulation is administered, the total dosage of 1500mg should be given as 500mg daily for 3 days. As an alternative, the same total dose can be given over 5 days with 500mg given on Day 1, then 250mg daily on Days 2 to 5.

Children:

The maximum recommended total dose for any treatment is 1500mg for children. Axel Azithromycin-250mg Tablet should only be administered to children weighing more than 45kg, with the same dosage used as in adult patients.

Special populations:

In the Elderly:

The same dosage as in adult patients is used in the elderly. Elderly patients may be more susceptible to the development of torsades de pointes arrhythmia than younger patients.

In Patients with Renal Impairment:

No dose adjustment is necessary in patients with mild to moderate renal impairment (GFR 10-80 ml/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR <10ml/min).

In Patients with Hepatic Impairment:

The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

CONTRAINDICATION:

The use of this product is contraindicated in patients with hypersensitivity to azithromycin, erythromycin, any macrolide or ketolide antibiotic.

PRECAUTION:

Hypersensitivity:

As with erythromycin and other macrolides, rare serious allergic reactions including angioneurotic oedema and anaphylaxis (rarely fatal), dermatologic reactions including acute generalised exanthematous pustulosis (AGEP), Stevens Johnson syndrome (SJS), toxic epidermal necrolysis (TEN) (rarely fatal) and drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment. If an allergic reaction occurs, the medicinal product should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued. In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) (e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalized exanthematous pustulosis (AGEP)), Axel Azithromycin-250mg Tablet should be discontinued immediately and appropriate treatment should be urgently initiated.

Hepatic impairment:

Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease. Cases of fulminant hepatitis potentially leading to life-threatening liver failure have been reported with azithromycin. Some patients may have had pre-existing hepatic disease or may have been taking other hepatotoxic medicinal products. In case of signs and symptoms of liver dysfunction, such as rapid developing asthenia associated with jaundice, dark urine, bleeding tendency or

hepatic encephalopathy, liver function tests / investigations should be performed immediately. Azithromycin administration should be stopped if liver dysfunction has emerged.

Ergot alkaloids and azithromycin:

In patients receiving ergotamine derivatives, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergotamine derivatives and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administered.

Superinfections:

As with any antibiotic preparation, observation for signs of superinfection with non-susceptible organisms, including fungi is recommended.

Clostridium difficile associated diarrhoea:

Clostridium difficile associated diarrhoea (CDAD) has been reported with the use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*. *C. difficile* produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antimicrobial agents. In case of CDAD anti-peristaltics are contraindicated.

Renal impairment:

In patients with severe renal impairment (GFR < 10 ml/min) a 33% increase in systemic exposure to azithromycin was observed.

Prolongation of the QT interval:

Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation
 - Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, cisapride and terfenadine; antipsychotic agents; antidepressants; and fluoroquinolones.
 - Patients with electrolyte disturbance, particularly in cases of hypokalaemia and hypomagnesaemia
 - Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
 - Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval.
- Epidemiological studies investigating the risk of adverse cardiovascular outcomes with macrolides have shown variable results. Some observational studies have identified a rare short term risk of arrhythmia, myocardial infarction and cardiovascular mortality associated with macrolides including azithromycin. Consideration of these findings should be balanced with treatment benefits when prescribing azithromycin.

Myasthenia gravis:

Exacerbations of the symptoms of myasthenia gravis and new onset of myasthenia syndrome have been reported in patients receiving azithromycin therapy.

Paediatric population:

Safety and efficacy for the prevention or treatment of *Mycobacterium avium* complex in children have not been established.

Infantile hypertrophic pyloric stenosis:

Infantile hypertrophic pyloric stenosis (HPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and / or inappetence with feeding occurs.

The following should be considered before prescribing azithromycin:

Azithromycin film-coated tablets are not suitable for treatment of severe infections where a high concentration of the antibiotic in the blood is rapidly needed. Azithromycin is not the first choice for the empiric treatment of infections in areas where the prevalence of resistant isolates is 10% or more. In areas with a high incidence of erythromycin A resistance, it is especially important to take into consideration the evolution of the pattern of susceptibility to azithromycin and other antibiotics. As for other macrolides, high resistance rates of *Streptococcus pneumoniae* (> 30%) have been reported for azithromycin in some European countries. This should be taken into account when treating infections caused by *Streptococcus pneumoniae*.

Pharyngitis / tonsillitis:

Azithromycin is not the substance of first choice for the treatment of pharyngitis and tonsillitis caused by *Streptococcus pyogenes*. For this and for the prophylaxis of acute rheumatic fever penicillin is the treatment of first choice.

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Sinusitis:

Often, azithromycin is not the substance of first choice for the treatment of sinusitis.

Acute otitis media:

Often, azithromycin is not the substance of first choice for the treatment of acute otitis media.

Skin and soft tissue infections:

The main causative agent of soft tissue infections, *Staphylococcus aureus*, is frequently resistant to azithromycin. Therefore, susceptibility testing is considered a precondition for treatment of soft tissue infections with azithromycin.

Infected burn wounds:

Azithromycin is not indicated for the treatment of infected burn wounds.

Sexually transmitted disease:

In case of sexually transmitted diseases a concomitant infection by *T. pallidum* should be excluded.

Neurological or psychiatric diseases:

Azithromycin should be used with caution in patients with neurological or psychiatric disorders.

Sodium:

Azithromycin Tablets contains less than 1 mmol sodium (23mg) per dose, that is to say essentially 'sodium-free'.

Effects on ability to drive and use machines:

There is no evidence to suggest that azithromycin may have an effect on a patient's ability to drive or operate machinery. Visual impairment and vision blurred may have an effect on a patient's ability to drive or operate machinery.

DRUG INTERACTIONS:

Antacids: No effect on overall bioavailability was seen, although the peak serum concentrations were reduced by approximately 25%. In patients receiving both azithromycin and antacids, the medicinal products should not be taken simultaneously. Azithromycin must be taken at least 1 hour before or 2 hours after the antacids. Co-administration of a 600mg single dose of azithromycin and 400mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Cetirizine: Coadministration of a 5-day regimen of azithromycin with cetirizine 20mg at steady-state resulted in no pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Didoxorynosine): Coadministration of 1200mg/day azithromycin with 400mg/day didanosine in 6 HIV-positive subjects did not appear to affect the steady-state pharmacokinetics of didanosine as compared with placebo.

Digoxin and colchicine: Concomitant administration of macrolide antibiotics, including azithromycin, with P-glycoprotein substrates, such as digoxin and colchicine, has been reported to result in increased serum levels of the P-glycoprotein substrate. Therefore, if azithromycin and P-glycoprotein substrates, such as digoxin are administered concomitantly, the possibility of elevated serum digoxin concentrations should be considered. Clinical monitoring, and possibly serum digoxin levels, during treatment with azithromycin and after its discontinuation are necessary.

Zidovudine: Single 1000mg doses and multiple 1200mg or 600mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients. Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergotamine derivatives: Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Pharmacokinetic studies have been conducted between azithromycin and the following drugs known to undergo significant cytochrome P450-mediated metabolism:

Atorvastatin: Coadministration of atorvastatin (10mg daily) and azithromycin (500mg daily) did not alter the plasma concentrations of atorvastatin (based on a HMG CoA-reductase inhibition assay). However, post-marketing cases of rhabdomyolysis in patients receiving azithromycin with statins have been reported.

Carbamazepine: No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine: In a pharmacokinetic study investigating the effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants: There have been reports received in the post-marketing period of potentiated anticoagulation subsequent to co-administration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin: In a pharmacokinetic study with healthy volunteers that were administered a 500mg/day oral dose of azithromycin for 3 days and were then administered a single 10mg/kg oral dose of cyclosporin, the resulting cyclosporin C_{max} and AUC₀₋₅ were found to be significantly elevated. Consequently, caution should be exercised before considering concurrent administration of these drugs. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz: Co-administration of a 600mg single dose of azithromycin and 400mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole: Co-administration of a single dose of 1200mg azithromycin did not alter the pharmacokinetics of a single dose of 800mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the coadministration of fluconazole, however, a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir: Coadministration of a single dose of 1200mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800mg three times daily for 5 days.

Methylprednisolone: Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam: Coadministration of azithromycin 500mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15mg dose of midazolam.

Nelfinavir: Co-administration of azithromycin (1200mg) and nelfinavir at steady state (750mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin: Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil: There was no evidence of an effect of azithromycin (500mg daily for 3 days) on the AUC and C_{max} of sildenafil or its major circulating metabolite.

Terfenadine: Pharmacokinetic studies have reported no evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however there was no specific evidence that such an interaction had occurred.

Theophylline: There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are coadministered to healthy volunteers. As interactions of other macrolides with theophylline have been reported, alertness to signs that indicate a rise in theophylline levels is advised.

Triazolam: Coadministration of azithromycin 500mg on Day 1 and 250mg on Day 2 with 0.125mg triazolam on Day 2 had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim / sulfamethoxazole: Coadministration of trimethoprim/sulfamethoxazole DS (160mg/800 mg) for 7 days with azithromycin 1200mg on Day 7 had no significant effect on peak concentrations total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

Astemizole, alfentanil: There are no known data on interactions with astemizole or alfentanil. Caution is advised in the co-administration of these medicines with azithromycin because of the known enhancing effect of these medicines when used concurrently with the macrolid antibiotic erythromycin.

Cisapride: Cisapride is metabolized in the liver by the enzyme CYP 3A4. Because macrolides inhibit this enzyme, concomitant administration of cisapride may cause the increase of QT interval prolongation, ventricular arrhythmias and torsades de pointes.

PREGNANCY AND LACTATION:

There are no adequate data from the use of azithromycin in pregnant women. In reproduction toxicity studies in animals azithromycin was shown to pass the placenta, but no teratogenic effects were observed. The safety of azithromycin has not been confirmed with regard to the use of the active substance during pregnancy. Therefore, azithromycin should only be used during pregnancy if the benefit outweighs the risk. Azithromycin has been reported to be secreted into human breast milk, but there are no adequate and well-controlled clinical studies in nursing women that have characterized the pharmacokinetics of azithromycin excretion into human breast milk. Because it is not known whether azithromycin may have adverse effects on the breast-fed infant, nursing should be discontinued during treatment with azithromycin. Among other things diarrhoea, fungus infection of the mucous membrane as well as sensitisation is possible in the nursed infant. It is recommended to discard the milk during treatment and up until 2 days after discontinuation of treatment. Nursing may be resumed thereafter.

SIDE EFFECTS:

Lists of the adverse reactions identified by system organ class and frequency. The frequency grouping is defined using the following convention: Very common; common; uncommon; rare; very rare; not known. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Infections and infestations: **Uncommon:** Candidiasis, Vaginal infection, Pneumonia, Fungal infection, Bacterial infection, Pharyngitis, Gastroenteritis, Respiratory disorder, Rhinitis, Oral candidiasis;

Not known: Pseudomembranous colitis

Blood and lymphatic system disorders: **Uncommon:** Leukopenia, Neutropenia, Eosinophilia;

Not known: Thrombocytopenia, Haemolytic anaemia

Immune system disorders: **Uncommon:** Angioedema, Hypersensitivity;

Not known: Anaphylactic reaction

Metabolic and nutrition disorders: **Uncommon:** Anorexia

Psychiatric disorders: **Uncommon:** Nervousness, Insomnia; **Rare:** Agitation, Depersonalisation;

Not known: Aggression, Anxiety, Delirium, Hallucination

Nervous system disorders: **Common:** Headache; **Uncommon:** Dizziness, Somnolence, Dysgeusia, Paraesthesia; **Not known:** Syncope, Convulsion, Hypoaesthesia, Psychomotor hyperactivity, Anosmia, Ageusia, Parosmia, Myasthenia gravis

Eye disorders: **Uncommon:** Visual impairment; **Not known:** Blurred vision

Ear and labyrinth disorders: **Uncommon:** Ear disorder, Vertigo;

Not known: Hearing impairment including deafness and / or tinnitus

Cardiac disorders: **Uncommon:** Palpitations; **Not known:** Torsades de pointes, Arrhythmia including ventricular tachycardia electrocardiogram QT prolonged

Vascular disorders: **Uncommon:** Hot flush; **Not known:** Hypotension

Respiratory, thoracic and mediastinal disorders: **Uncommon:** Dyspnoea, Epistaxis

Gastrointestinal disorders: **Very common:** Diarrhoea; **Common:** Vomiting, Abdominal pain, Nausea

Uncommon: Constipation, Flatulence, Dyspepsia, Gastritis, Dysphagia, Abdominal distension, Dry mouth, Eructation, Mouth ulceration, Salivary hypersecretion; **Not known:** Pancreatitis, Tongue discolouration.

Hepatobiliary disorders: **Uncommon:** Hepatitis; **Rare:** Hepatic function abnormal, Jaundice cholestatic **Not known:** Hepatic failure (which has rarely resulted in death), Hepatitis fulminant, Hepatic necrosis

Skin and subcutaneous tissue Disorders: **Uncommon:** Rash, Pruritus, Urticaria, Dermatitis, Dry skin, Hyperhidrosis; **Rare:** Photosensitivity reaction, Acute generalised exanthematous pustulosis (AGEP)

Not known: Erythema multiforme, severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)

Musculoskeletal and connective tissue disorders: **Uncommon:** Osteoarthritis, Myalgia, Back pain, Neck pain; **Not known:** Arthralgia

Renal and urinary disorders: **Uncommon:** Dysuria, Renal pain; **Not known:** Renal failure acute, Nephritis interstitial

Reproductive system and breast disorders: **Uncommon:** Metrorrhagia, Testicular disorder

General disorders and administration site conditions: **Uncommon:** Oedema, Asthenia, Malaise, Fatigue, Face oedema, Chest pain, Pirexia, Pain, Peripheral oedema

Investigations: **Common:** Lymphocyte count decreased, Eosinophil count increased, Blood bicarbonate decreased, Basophilis increased, Neutrophils increased; **Uncommon:** Aspartate aminotransferase increased, Alanine aminotransferase increased, Blood bilirubin increased, Blood urea increased, Blood creatinine increased, Blood potassium abnormal, Blood alkaline phosphatase increased, Chloride increased, Glucose increased, Platelets increased, Hematocrit decreased, Bicarbonate increased, Abnormal sodium

Injury and poisoning: **Uncommon:** Post procedural complication

Adverse reactions possibly or probably related to *Mycobacterium Avium* Complex prophylaxis and treatment. These adverse reactions differ from those reported with immediate release or the prolonged release formulations, either in kind or in frequency.

Metabolism and nutrition disorders: **Common:** Anorexia

Nervous system disorders: **Common:** Dizziness, Headache, Paraesthesia, Dysgeusia;

Uncommon: Hypoaesthesia

Eye disorders: **Common:** Visual impairment

Ear and labyrinth disorders: **Common:** Deafness; **Uncommon:** Hearing impaired, Tinnitus

Cardiac disorders: **Uncommon:** Palpitations;

Gastrointestinal disorders: **Very common:** Diarrhoea, Abdominal pain, Nausea, Flatulence, Abdominal discomfort, Loose stools

Hepatobiliary disorders: **Uncommon:** Hepatitis

Skin and subcutaneous tissue disorders: **Common:** Rash, Pruritus; **Uncommon:** Steven-Johnson syndrome, Photosensitivity reaction

Musculoskeletal and connective tissue disorders: **Common:** Arthralgia

General disorders and administration site conditions: **Common:** Fatigue;

Uncommon: Asthenia, Malaise

Postmarketing Experience:

Cardiac Disorders: Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes.

Gastrointestinal Disorders: Infantile hypertrophic pyloric stenosis.

OVERDOSAGE AND TREATMENT:

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdose, general symptomatic and supportive measures are indicated as required.

STORAGE:

Store below 30°C. Protect from light.

KEEP OUT OF REACH OF CHILDREN

JAUHI DAR KANAK-KANAK

PACK QUANTITIES

Available in blister pack of 1 x 6's.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder & Manufactured By:

 **Kotra Pharma (M) Sdn. Bhd.**
No. 1, 2 & 3, Jalan TTC 12, (90002-4)
Cheng Industrial Estate,
75250 Melaka, Malaysia.